

SHIVANSH PRATAP SINGH

Software Engineer

@ engrshivansh@gmail.com

9560359771

engrshivansh

Shiv07ansh

PROFESSIONAL SUMMARY

Software Engineer with strong C/C++ systems programming experience in EDA and embedded-scale software, specializing in performance-critical pipelines, complex data models, and cross-platform development (Linux/Windows). Hands-on experience owning production features, debugging architectural issues, and delivering releases under tight timelines. Strong CS fundamentals, data structures, and low-level system understanding.

EXPERIENCE

Cadence Design Systems

Feb 2024

Noida

C++ Systems Feature Development (EDA Tooling):

- Owned end-to-end development of Symbol Editor features for Allegro OrCAD X/DEHDL, working on large scale, performance critical codebases.
- Designed and optimized core data structures, traversal algorithms, and metadata caching layers, significantly reducing symbol load latency and improving UI responsiveness.
- Implemented robust backend logic integrated with Qt-based UI, ensuring consistency across multiple editor modes and workflows.

Architecture, APIs Cross-Team Collaboration:

- Extended and debugged REST APIs enabling reliable backend- UI data exchange for library management systems.
- Collaborated with architects, QA, and product teams to evaluate new features, resolve design trade-offs, and unblock customer-critical workflows.
- Reviewed and refined feature designs, menu behaviors, and interaction flows under aggressive release timelines.

Performance Debugging Tooling:

- Diagnosed and resolved high-impact bugs by deep-diving into symbol / component data-flow models.
- Worked extensively in Linux environments, using internal CI/CD pipelines, Perforce, and regression testing frameworks.
- Authored unit and regression tests in Tcl, validating both new and refactored functionality.

Key Achievements:

- Developed and shipped new features for Allegro's DEHDL symbol editor, included in release 24.1.
- Took ownership of a full feature-set and its APIs; collaborated with QA and design teams to debug and integrate.
- Implemented critical menu-behavior fixes and UI flow enhancements under aggressive timelines to unblock key customer workflows.

Datacouch.io

Nov 2022 – March 2023

Noida

- Individual AI-ML project on Face mask/glasses detection in Python.
- Created automated Zoom attendance system for seminars using Webhooks and Zapier APIs.
- Created consulting presentations for various cloud storage solutions (AWS, S3, Azure, Snowflake, GCP) and a blog on Apache Airflow.
- Designed a tutorial for various visualizations using Power BI with real customer data.

PROJECTS

AIoT Medicine Reminder System

Primary Author | Springer LNNS (ICDECT 25)

- Designed and deployed an edge-AI embedded system on ESP32-S3, integrating TTS, alarms, modular extensions, and local inference.
- Built a voice-driven intent-classification pipeline, optimized for memory and energy constraints.
- Designed system architecture with clear modular separation for future scalability.

Cascading Neural Network model for Efficient Speech Detection

- Designed a two-stage inference pipeline combining lightweight KWS (Keyword Spotting) with SLU (Spoken Language Understanding) for energy-efficient command recognition.
- Applied quantization and pruning to optimize runtime memory footprint.

Disease Detection in Plants

- Implemented CNN-based classification using TensorFlow/Keras.
- Focused on dataset preparation, evaluation, and performance tuning.

TRAINING & COURSES

- C++ For C Programmers (A & B)** -University of California [Coursera, 2024]
- Data Structures** - University of California San Diego [Coursera, 2024]
- Digital Systems: From Logic Gates to processors** - Universitat Autònoma de Barcelona [Coursera, 2024]
- VLSI CAD Part I: Logic** - University of Illinois Urbana-Champaign [Coursera, 2024]
- Linear Circuits 2: AC Analysis** - Georgia Institute of Technology [Coursera, 2024]

EDUCATION

B.Tech (Hons.) in Computer Science

UPES Dehradun

August 2019 – Dec 2023

Class XII

Delhi Public School, Greater Noida

April 2016 – June 2017